Auditing GBFS bike-sharing feeds: a reproducible anomaly taxonomy for open mobility data, validated in France and projected across 48 countries

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Abstract

Open mobility feeds are increasingly assumed to be directly ingestible into quantitative research pipelines. The General Bikeshare Feed Specification (GBFS) is now mandatory for French operators and is catalogued worldwide by MobilityData, but syntactic standardisation does not guarantee the semantic viability that research reuse requires. We audit the 123 French GBFS systems and project the rule set across the 1,509-system MobilityData catalogue (48 countries). From this we derive a data-quality taxonomy of seven classes (A1–A7): five *structural errors* that violate the implicit semantic contract of the field they populate, and two *semantic warnings* for spec-compliant patterns that make the `capacity` column non-aggregable. Detection runs as a nine-step idempotent and reversible purging protocol, and we release the *GBFS Audit Catalogue* (46,307 stations, 97 cities, 46 typed columns) on Zenodo under ODbL. The protocol removes 30.9 % of the raw French stations and relabels a further 61 %, so fewer than 9 % survive into the catalogue untouched. The downstream effect is large: correcting a single class (A3) drops Bordeaux from the top quartile to below the median of the national dock-count distribution, which measures the policy cost of skipping the audit. Projected at French-calibrated thresholds, the anti-patterns appear global rather than French-specific: 245 systems are flagged by A7 alone (97,314 stations), led by one operator that declares `capacity = NaN` across entire national deployments. A pre-registered, blind re-annotation of 422 stations confirms construct validity: 96.8 % of adjudicated A3 flags mark genuinely defective records, every adjudicated legacy-only geospatial flag is a confirmed false positive (14/14), and one adjudicated station in five is a ghost with no infrastructure on the ground. Any quantitative use of GBFS data should therefore be preceded by an explicit and reproducible audit. The pipeline and a Docker image are released for reproducible rebuilds.

Highlights

- Seven-class taxonomy (5 structural, 2 semantic); projected across 1,509 GBFS systems
- Topology-aware A4 detector reclassifies 8,005 legacy flags (pre-registered blind audit: 14/14 adjudicated flags false)
- Leave-one-operator-out CV on 7 operators supports rule generalisability
- 30.9 % of raw FR stations reclassified; blind audit: 1 in 5 stations is a ghost
- Released as a versioned Parquet (46k stations) with Docker and a hold-out test

Keywords: GBFS; data quality; open mobility data; FAIR; reproducibility; bike sharing; standards compliance.

1 Introduction

For more than a decade, bike-sharing has been a structural component of urban mobility policy in France and across Europe [1, 2, 3]. To support inter-operability between fleets, the General

Bikeshare Feed Specification (GBFS), introduced by the North American Bikeshare Association in 2015 and now maintained by the Open Mobility Foundation, provides a shared ontology for station geolocation, dock counts, real-time availability and pricing [4, 5]. In France, the 2019 Mobility Orientation Law [6] (LOM, article L. 1115-1 of the transport code) makes publishing these feeds on the national portal *transport.data.gouv.fr* [7] a legal obligation. By 2026, many research and policy pipelines assume that GBFS feeds can be ingested directly.

That assumption is misleading. Our audit of 123 French GBFS systems shows that standardised syntax hides substantial semantic heterogeneity. The seven classes of Section 2.2 (Table 1) separate five *structural errors*, which violate the implicit semantic contract of the field they populate, from two *semantic warnings*, which flag spec-compliant patterns whose interpretation by a downstream consumer is ambiguous. The `station_information.capacity` field alone covers radically different realities: a physical dock count for dock-based fleets, an estimator obtained by conditional averaging on free-floating fleets, and arbitrary placeholders for under-reporting operators. The ambiguity reaches the object of study itself: seventeen car-sharing systems (fourteen under the *Citiz* brand) are listed on the national portal under the GBFS schema. It persists even among the systems that look clean, where 1.1 % of the certified stations are flagged by A4 as 3σ geospatial outliers (down from 3.6 % under the legacy centroid detector) and four further systems publish bounding boxes above the 50,000 km^2 A5 threshold or list stations outside the national perimeter. The methodological conclusion is straightforward: standardising the syntax of an open feed is not enough to make it a research artefact. Any quantitative use of GBFS data requires an explicit and reproducible audit.

Empirical audits of mobility feeds have so far remained piecemeal. Bike-sharing demand modelling and rebalancing analysis have generated a rich operational literature [8, 9, 10], but all of these works treat the input feeds as given and do not audit the standard itself. The closest precedent in transit-data quality is the smart-card literature: Pelletier et al. [11] systematically catalogue the gaps between what fare-collection feeds nominally publish and what they operationally support for research reuse, a programme close in spirit to the GBFS audit we report here. Tao et al. [12] document sampling biases in mobile-phone records used for origin-destination estimation; Romanillos et al. [13] warn against statistical artefacts of ingesting raw GPS feeds directly; Bachand-Marleau et al. [14] surfaced operational-definition heterogeneity in bike-sharing demand data more than a decade ago; and Fishman [15] stresses the heterogeneity of operational definitions in international BSS comparisons. The North American Bikeshare and Scootershare Association tracks cross-operator harmonisation gaps annually [16], with the regulatory framing typical of an industry-side body but without the per-row audit trail a research consumer needs. In the adjacent transit literature, the GTFS community has been more proactive: the Canonical GTFS Schedule Validator [17] and the earlier Conveyal validator [18] codify hundreds of schema-level checks on transit feeds, and the data-engineering community has produced reusable declarative-validation frameworks such as Great Expectations [19], Pandera [20] and the Frictionless Data specifications [21]. The broader anomaly-detection literature [22] provides a taxonomy of detection paradigms (statistical, proximity-based, clustering-based) that contextualises the spatial outlier methods we adopt in Section 4.2. The underlying data-quality literature is mature, from the consumer-centred dimensions of Wang and Strong [23] through the assessment, governance and profiling frameworks that followed [24, 25, 26, 27] to the ISO/IEC 25012 and 25024 quality standards [28, 29], their geographic counterpart ISO 19157 [30] (made regulatory by the INSPIRE directive [31]), and the machine-readable DCAT and Data Quality Vocabulary of the W3C [32, 33]. None of these, however, addresses GBFS specifically or produces a versioned, audited reference dataset at the scale of a country. The present paper fills that gap for the French case and extends to the global GBFS catalogue.

This paper makes five contributions. First, a **taxonomy** of seven anomaly classes (A1–A7), derived from the audit of the 123 French GBFS systems and the 1,509-system MobilityData catalogue. It is stress-tested on a live panel of thirteen non-French systems spanning six station-management stacks (PBSC, JCDecaux Cyclocity, Smartbike, De Lijn / Blue-bike, Urban Sharing AS, Deutsche Bahn) across five countries; twelve serve as metropolitan negative controls and all pass classes A1, A2, A3, A6 and A7 under unchanged thresholds. Second, a nine-step **purging protocol** (Algorithm 1) that operationalises the seven classes and logs every transformation reproducibly and reversibly. Third, the **Audit Catalogue GBFS France** (46,307 certified stations, 97 cities), released as a versioned Parquet file with DOI `[deposited on Zenodo; DOI withheld for review]` under ODbL, with its JSON Schema, Croissant manifest and audit report, and mirrored as a Hugging Face dataset. Fourth, a reusable **reference implementation**: the A1–A7 detectors are released as `gbfs-toolkit`, an independently versioned (SemVer), `pip`-installable Python library archived on Zenodo under its own DOI (`[reference-implementation library archived on Zenodo; DOI withheld for review]`) and numerically cross-validated against independent estimators, with the catalogue's nine-step protocol (`audit_pipeline`, MIT) a thin domain-orchestration layer that calls it. Fifth, a **roadmap** of nine falsifiable experiments, six already with completed results, so that the protocol can be challenged in public rather than defended in private. Section 2 presents the taxonomy and the purging protocol, Section 3 reports the empirical impact on the French and global corpora, Section 4 the validation and robustness analyses, and Section 5 discusses implications and limitations.

2 Taxonomy and purging protocol

2.1 Provenance of the taxonomy

The seven-class taxonomy A1–A7 was built by a hybrid inductive-deductive workflow over six iterations between 2025-04 and 2026-04, starting from the four ISO/IEC 25012 data-quality dimensions (completeness, consistency, accuracy, currentness) and enriched by reverse engineering of both the French free-floating fleets and the global MobilityData catalogue. The five structural classes came first. A1 and A4 emerged from an exploratory scan of the 123 French systems, while A5 surfaced from enrichment with the *transport.data.gouv.fr* catalogue. Capacity diagnostics yielded A2 (a zero-variance signature on `capacity`) and A3 (a conditional-averaging estimator matching *Pony*'s per-system median capacity more closely than a physical-dock count). The two semantic warnings came later, from the global sweep: A6 from the empirical distribution of zero-capacity stations across the 1,509 globally audited systems (Section 2.3), and A7 from the bimodal 0 %/100 % distribution of NaN-capacity stations (Section 2.4). Two rejected hypotheses ("rolling-stock noise", collapsed into A3; "cross-border duplicate", no instance found) ship with the audit report for transparency.

2.2 The seven-class taxonomy

The audit of the 123 French GBFS systems, complemented by the global sweep of Section 3.4, defines seven recurring data-quality classes: five structural errors (A1–A5) and two semantic warnings (A6–A7), summarised in Table 1. Each class, in isolation, is sufficient to invalidate a non-audited cross-system comparison; the detection rules are formalised in Section 2.5.

Of the 142 French GBFS feeds at the national portal, 123 are certified into the Audit Catalogue, 2 fail technical ingestion, and the 17 car-sharing systems (A1) are filtered upstream.

A3 is the class with the largest downstream impact on the French corpus. Free-floating fleets (*Pony*, *Cykleo*, etc.) advertise virtual stations in GBFS, the term "station" designating the GPS position of an anchoring point currently occupied by a vehicle. To avoid displaying empty stations, the operator typically reports a capacity profile by *conditional averaging* on stations whose instantaneous capacity is non-zero:

$$\bar{c}_{\text{profile}} = \frac{\sum_{i\,:\,c_i>0} c_i}{\#\{i\,:\,c_i>0\}} \;\neq\; \bar{c}_{\text{actual}} = \frac{1}{N}\sum_{i=1}^{N} c_i. \tag{1}$$

Table 1: The unified GBFS data-quality taxonomy: five *structural errors* (A1–A5) that violate the implicit semantic contract of the field they populate, plus two *semantic warnings* (A6–A7) that flag spec-compliant publication choices whose downstream-consumer interpretation is nevertheless ambiguous. **Both columns are produced by the same library function** (`gbfs_toolkit.audit_static`, identical thresholds): *FR* is the incidence on the released, certified French catalogue (123 systems), and *World* is a single-pipeline re-audit of the MobilityData catalogue (936 systems that publish a populated `station_information`, 46 countries, France included), re-fetched live and audited by the same function (Section 3.4). The feed-intrinsic classes A2, A4, A5 and A7 are directly comparable; A1 (‡) is read from the GBFS v3 `vehicle_types form_factor`, not from an operator name. A3 (†) counts the free-floating share under the same broad definition as the FR column, with the over-capacity signature that isolates the Bordeaux mechanism reported separately in the footnote. Every count is reproducible: the World column from the frozen feed archive (deterministic, seeded; `run_unified_audit.py`), the FR column from the released parquet.

Class	Name	Signature	FR	World
Structural errors (semantic-contract violations)				
A1	Out-of-domain inclusion	Car-sharing advertised as BSS	17	43‡
A2	Placeholder capacity	Constant non-zero c_i across docked subset	1	8
A3	Structural over-capacity	Conditional averaging on free-floating	41	699†
A4	Geospatial error	3σ outlier on per-system nearest-neighbour distance	88	515
A5	Out of perimeter	System bounding box $> 50{,}000\,\mathrm{km}^2$	4	30
Semantic warnings (spec-compliant but consumer-ambiguous)				
A6	Zero-capacity dock	$\geq 1\,\%$ of docked stations declare $c = 0$	0	0
A7	Null capacity field	$\geq 50\,\%$ of stations declare $c =$ `NaN`	32	288

† A3 counts the free-floating share (natively declared or reclassified by S3), the same broad definition as the FR column. The over-capacity reclassification ($\bar{c}_{\text{profile}}/\bar{c}_{\text{actual}} > 5$) is computed from `station_information` alone by the library's `overcapacity_ratio`; 34 of the 699 World free-floating systems carry that over-capacity signature, reproducing the 33 found in the frozen snapshot.

‡ A1 is detected feed-intrinsically from the GBFS v3 `vehicle_types form_factor`: a system declaring a `car` form factor (and no plain `bicycle`) is car-sharing, read straight from the feed rather than from any operator name. Of the 936 audited systems, 876 publish `vehicle_types`, and 43 declare a car fleet; 16 of these are French, matching the catalogue's 17 hand-checked car-share systems. The exact-enum match avoids the `cargo_bicycle` false positive. An operator-name heuristic is needed only for the $\sim 6\,\%$ of feeds without `vehicle_types`.

Aggregated at the system level, $\bar{c}_{\text{profile}}$ wrongly classifies thousands of free-floating bikes as heavy dock-based stations. Cross-system comparison is therefore meaningful only after explicit reclassification; the empirical signature is visible in Figure 1 as a bimodal capacity distribution.

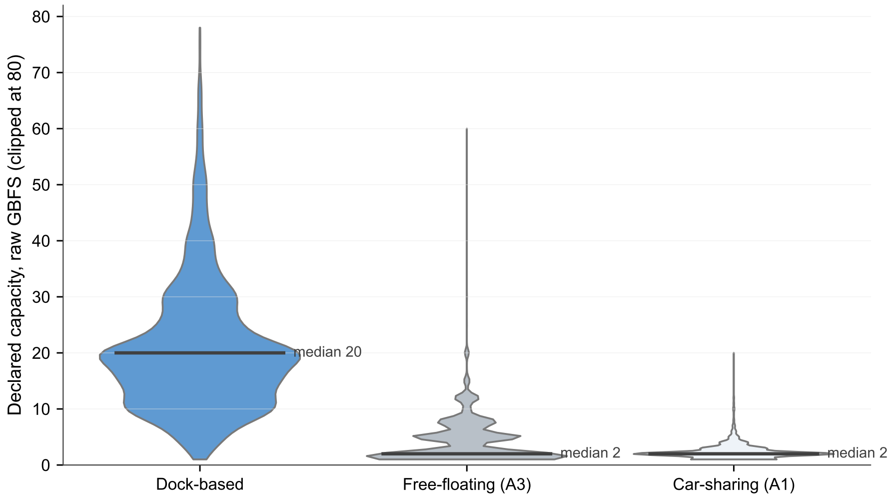

Figure 1: Empirical signature of the A3 floating-anchor bias. Declared capacity distribution by `station_type` (clipped at 80 docks) on the Audit Catalogue corpus. Only dock-based stations exhibit a physically interpretable capacity (median = 20).

2.3 A6: zero-capacity dock formalisation

The global scalability projection of Section 3.4 surfaced a recurring pattern not covered by A1–A5: a non-trivial fraction of stations in `station_information` declare `capacity = 0`. Because the GBFS schema reserves `station_information` for physical infrastructure and provides `station_status` for transient counts, a non-virtual physical station with zero docks is semantically inconsistent. The signature is therefore:

$$r_s^{A6} = \frac{\#\{i \in \mathcal{S}_s \,:\, c_i = 0 \wedge \texttt{is\_virtual\_station}(i) \neq \texttt{true}\}}{|\mathcal{S}_s|} \geq \tau_{A6}. \tag{2}$$

Threshold τ_{A6} and the empirical distribution. A6 is evaluated on the docked systems of the global sweep, where a zero-capacity station is semantically inconsistent; on free-floating fleets $c = 0$ is expected and carries no information, so they are excluded from the base. Across the 301 docked systems with at least 20 stations (Table S2), 89.0 % declare *exactly zero* stations with $c = 0$, establishing the dock-based baseline, while the remaining 11.0 % form a heavy tail with mean $\bar{r} = 0.36\,\%$ and standard deviation $\hat{\sigma} = 2.41\,\%$. The protocol threshold $\tau_{A6} = 1\,\%$ (Algorithm 1, step S7) isolates the upper tail: 22 docked systems globally cross it. The threshold is deliberately conservative, so that a flagged system declares zero-capacity on a non-trivial share of its docks rather than on the odd mis-entered station.

The empirical distribution of r_s^{A6} and the per-class incidence on the audited corpus are reported in Section 3.5 (Results, Table S2).

2.4 A7 (semantic warning): null capacity field formalisation

The Italian case study of Section 3.4 demonstrates that A1–A6 are structurally blind to operators that populate `station_information` but leave the `capacity` field unfilled. Like A6, A7 is by design a consumer-side semantic warning rather than a publisher-side anomaly. `capacity = NaN` is explicitly permitted by the GBFS v3 specification, in particular for free-floating fleets where the field has no physical referent; a system that publishes `capacity = NaN` on every entry is *spec-compliant*. What A7 captures is the downstream consequence: any analytical pipeline that sums or averages the `capacity` column across such a system aggregates NaN with non-NaN values silently, and produces a number whose physical meaning is undefined. The flag is therefore tooling for the consumer ("do not arithmetically combine these stations with others"), not a verdict on the publisher. The empirical signature is:

$$\rho_s^{A7} = \frac{\#\{i \in \mathcal{S}_s : c_i = \texttt{NaN}\}}{|\mathcal{S}_s|} \geq \tau_{A7}, \tag{3}$$

with the conservative threshold $\tau_{A7} = 0.5$ chosen so that a system flagged by A7 declares no usable capacity on at least half of its entries. Unlike A6, ρ_s^{A7} is extremely bimodal on the global corpus (Table S3): 87.0 % of audited systems sit at one of the two extreme values 0 % or 100 % NaN, with only 13.0 % distributed across the interior. Any threshold in $(0, 1)$ partitioning the interior in the same way yields essentially the same flagged set; $\tau_{A7} = 0.5$ is reported for parsimony and is robust to sensitivity sweeps in the tested range.

The empirical distribution of ρ_s^{A7} and the per-class incidence on the audited corpus are reported in Section 3.5 (Results, Table S3).

2.5 The nine-step purging protocol

The transition from the raw feed to the Audit Catalogue proceeds through a chain of nine sequential transformations (Algorithm 1), each coded in `utils/data_loader.py` and logged in an audit report shipped alongside the dataset. The protocol is deliberately sequential: early steps are inexpensive and remove out-of-domain or coarse-grained anomalies (A1, A2, A4 perimeter), while later steps require system-level statistics ($\bar{c}$, centroids, σ) and therefore benefit from operating on a pre-cleaned set.

2.6 Protocol properties

Four properties are designed into the pipeline. *Idempotence:* two successive applications produce the same result, tested by a unit test executed on every CI run. *Reversibility:* each excluded station is kept in `rejected_stations.parquet` together with the exclusion motive. *Logging:* every step records the number of stations in, the number of stations out and the type of transformation applied. *Explicit parameterisation:* all thresholds ($\sigma_{\max}$, $N_{\min}$, geofilter bounds, hybridisation radii) are documented as hyperparameters and can be varied for sensitivity analysis without modifying the source code.

These properties are not arbitrary engineering choices: they operationalise the ISO/IEC 25012 / 25024 data-quality dimensions [28, 29] at the domain level – idempotence for internal consistency, the `rejected_stations.parquet` sidecar for recoverability and traceability, step-level logging for traceability, explicit hyperparameters for understandability, and the FAIR release (Section 2.10) for compliance (full mapping in Table S9). We are not aware of a published reference implementation of ISO/IEC 25012 on an open mobility standard prior to this work.

2.7 Threshold justification

The topological threshold $\sigma_{\max} = 3$ is a compromise between robustness to GPS noise and the preservation of legitimate peripheral extensions, *e.g.*an off-centre TGV station or a university campus. Section 4.4 reports the single-dimension sensitivity pilot: Top-10 invariance for $\sigma_{\max} \geq 2.5$. The robustness threshold $N_{\min} = 20$ dock-based stations is calibrated on the literature of meshed networks: below this value, density and connectivity metrics become noise-dominated and are no longer informative for cross-city comparison [34]. The A5 macro-region threshold of 50,000 km^2 is calibrated at the order of magnitude of a French métropolitaine *région administrative*

Input: National catalogue $\mathcal{C}$ (123 systems); thresholds $\sigma_{\max} = 3$, $N_{\min} = 20$, $\tau_{A2}^{\text{size}} = 20$, $\tau_{A6} = 0.01$, $\tau_{A7} = 0.5$, perimeter $\Phi = [41°, 52°] \times [-6°, 10°]$.

Output: Certified set $\mathcal{G}$, rejected set $\mathcal{R}$, seven per-row anomaly flags $f_{A1} \dots f_{A7}$, audit report.

$\mathcal{G} \leftarrow \emptyset$; $\mathcal{R} \leftarrow \emptyset$

foreach *system* $s \in \mathcal{C}$ **do**

- Ingest GBFS feed $\to \mathcal{S}_s$
- **S1.** If s is car-sharing, relabel `carsharing` and set $f_{A1} = \texttt{True}$
- **S2.** If $\mathrm{Var}(c) = 0$, $c > 0$ and $|\mathcal{S}_s^{\text{dock}}| \geq \tau_{A2}^{\text{size}}$, set $f_{A2} = \texttt{True}$ and force `free_floating`
- **S3.** Compute $\bar{c}_{\text{profile}}/\bar{c}_{\text{actual}}$; if > 5.0 relabel `free_floating` and set $f_{A3} = \texttt{True}$
- **S4.** Keep $\mathbf{x}_i \in \Phi$; mark stations outside as out-of-perimeter
- **S5.** Compute composite anomaly score $a_i = \alpha\, \hat{o}_i + (1 - \alpha)\, s_i$; if $a_i > 0.8$ set $f_{A4} = \texttt{True}$ (Section 4.2); fallback: nearest-neighbour robust 3σ envelope
- **S6.** Compute system bbox area A_s; if $A_s > 50{,}000\,\text{km}^2$, set $f_{A5} = \texttt{True}$ on every $i \in \mathcal{S}_s$
- **S7.** If $\#\{i : c_i = 0 \wedge \text{docked}\}/|\mathcal{S}_s| \geq \tau_{A6}$, set $f_{A6} = \texttt{True}$ on every $i \in \mathcal{S}_s$
- **S8.** If $\#\{i : c_i = \texttt{NaN}\}/|\mathcal{S}_s| \geq \tau_{A7}$, set $f_{A7} = \texttt{True}$ on every $i \in \mathcal{S}_s$
- **S9.** If $|\mathcal{S}_s^{\text{dock}}| < N_{\min}$, label `too_small` and route to $\mathcal{R}$
- Update $\mathcal{G}, \mathcal{R}$ and the per-row flag matrix accordingly

end

return $(\mathcal{G}, \mathcal{R}, \{f_{A1} \dots f_{A7}\}, \text{audit report}, \text{metadata})$

Algorithm 1: GBFS raw-to-Audit-Catalogue purging pipeline. Steps S1–S3 both purge and label (a station relabelled to `free_floating` or `carsharing` stays in the catalogue with its A-flags set); S4 implements the national perimeter filter; S5 the A4 topology-aware outlier check (Section 4.2); S6 the A5 macro-region surface check; S7–S8 the A6 / A7 capacity-field warnings. S9 is the only truly-removing step.

(median area ≈ 47,000 km^2 after the 2016 reform: Bretagne 27,000, Pays de la Loire 32,000, Centre-Val de Loire 39,000, Hauts-de-France 31,000, Grand Est 57,000, Auvergne-Rhône-Alpes 69,000, Occitanie 72,000). A single GBFS feed whose bounding box exceeds this scale plausibly aggregates operationally-independent deployments and the resulting system-level statistics (centroid, intra-system KNN, capacity profile) are no longer geographically interpretable.

2.8 Multi-source hybridisation

The output of the purging stage enriches each station with a contextual profile obtained by articulating six reference sources (Table S5). Hybridisation is performed either by spatial joining (point-in-polygon, buffer queries) or by administrative aggregation (*i.e.*at the IRIS or municipal level), depending on the granularity of the source. Each row of Table S5 corresponds to one or more typed columns of the released parquet schema (Table S4).

2.9 Released schema

The Audit Catalogue Parquet exposes 46 columns documented in Table S4: 14 from the raw GBFS payload (preserved verbatim or normalised), 11 from the audit pipeline (typed `station_type` enum, capacity-raw and capacity-audited pair, seven per-station anomaly flags A1–A7, operator label, and audit-confidence rating), 5 from intra- and inter-system network geometry, 4 from spatial-equity hybridisation, 3 from infrastructure hybridisation, and 2 from multimodal-access hybridisation. Every column carries a stable name, a declared dtype, a source pointer and a completeness rate; the machine-readable equivalent is shipped as a JSON Schema, a Frictionless Data Package descriptor and a Croissant manifest. The eleven audit-pipeline columns are the key novelty of the catalogue: they make the audit's verdict inspectable per row, rather than aggregated in a separate report.

Empirical completeness on the 5,442 dock-based stations is above 93 % for every contextual variable except the optional `address` field of the raw GBFS payload (Figure S4). The minimum-completeness threshold of 90 % documented in the audit report is therefore reached on every certified column of the released schema.

2.10 Reproducibility infrastructure

The Audit Catalogue is released in alignment with the four FAIR principles [35]. *Findable:* primary deposit on Zenodo under concept DOI `[deposited on Zenodo; DOI withheld for review]`, with one DOI per SemVer-tagged release (current 1.0.0, dated 2026-05); mirrored on *Recherche Data Gouv* (HAL/BSO indexing). *Accessible:* Parquet format, direct HTTPS download and Git LFS mirror. *Interoperable:* the schema extends GBFS v3 and is published as a JSON Schema, a Frictionless Data Package descriptor [21], a DCAT-AP record [36] and a Croissant manifest [37]. *Reusable:* released under ODbL v1.0; code under MIT, with audit report and source code. The A1–A7 detectors are additionally distributed as the `gbfs-toolkit` library (`pip install gbfs-toolkit`), versioned under SemVer and archived on Zenodo under its own DOI (`[reference-implementation library archived on Zenodo; DOI withheld for review]`), so a downstream study can pin the exact detector version it relied on; the human-validation gold standard (`gold_labels.csv`), the inter-rater reliability report and the blind annotation tool ship with the repository (`experiments/human_validation/`). Raw sources are archived with SHA-256 hashes; the Python environment is declared in `requirements.txt` (with `gbfs-toolkit` pinned to the exact release used to certify the published Parquet); a Docker image is published for reproducible rebuilds. The audit pipeline itself is fully deterministic (no random sampling); the only stochastic step is the bootstrap resampling used for the confidence intervals reported in the main text, for which the random seed is pinned at `numpy.random.default_rng(42)`. Idempotence and per-class detector unit tests run in CI on every push; the audit report is regenerated automatically on every release tag.

2.11 Implementation

The seven-class taxonomy is implemented as `gbfs-toolkit`, a research-grade, `pip`-installable Python library (`numpy`/`scipy`/`pandas` core) that exposes the A1–A7 detectors as pure, version-independent functions over a canonical station frame. The catalogue's nine-step protocol is a thin domain-orchestration package (`audit_pipeline`, MIT) that calls `gbfs_toolkit.audit_static` and adds the catalogue-specific hybridisation; the A1–A7 verdicts themselves come from the library. Calling it with `a7_scope="all"` and the catalogue's `a4_sigma` reproduces the original A7 definition (a fully-NaN-capacity system) and the A4 sensitivity exactly, so the released 46,307-station Parquet is regenerated bit-for-bit from the library and loads in under a second into a standard `pandas` session.

Software validation. Releasing the detectors as a standalone library lets them be validated to a standard a notebook pipeline rarely reaches. `gbfs-toolkit` carries 213 tests run on a continuous-integration matrix (Linux/Windows/macOS $\times$ Python 3.10–3.13, with static type checking and example-notebook execution), and validates its spatial and survival estimators *numerically against independent reference implementations*: global Moran's I and the LISA local indicators are cross-checked against `esda/PySAL` and the outage-survival curves against `lifelines`' Kaplan–Meier, to within numerical tolerance. The A1–A7 detectors themselves are pinned by deterministic per-class synthetic fixtures engineered to trip exactly one class (e.g. a 25-station grid with one transposed-coordinate row for A4, a 32-station fleet spanning more than 300,000 km^2 for A5, a 60 %-NaN-capacity dock-based system for A7), by specificity controls verifying that healthy fixtures stay unflagged, by property-based tests (`hypothesis`) asserting the mathematical invariants of each estimator on randomly generated feeds, and by a golden-master regression that freezes the end-to-end verdicts so any change of behaviour is caught. The suite further checks that the Tier-2 geometry columns (intra- and inter-system k-nearest-neighbour distances, local densities) are consistent with the equirectangular projection used for the meter-scale arithmetic, and that the audit is idempotent under repeated application. Tests run as a GitHub Actions workflow on every push and pull request, and locally inside the published Docker image. This validation establishes the correspondence between the implemented logic and the published anomaly taxonomy; it does not substitute for the hand-curated cross-operator audit and the author-led consistency check (Section 4.5) against which the verdicts were calibrated.

3 Empirical results

3.1 Cumulative effect on the national corpus

Table 2 summarises the cumulative effect of the purging protocol. From 67,000 raw `station_information` entries published by the 142 inventoried French GBFS feeds, 20,693 (30.9 %) are removed from the catalogue as falling outside the certification criteria (excluded micro-networks below N_{min}, technical-ingestion failures, perimeter and outlier filters); the remaining 46,307 stations are kept in the catalogue with an explicit `station_type` and per-row anomaly flags, of which a further 40,865 (61.0 % of the raw) are *relabelled* away from their declared GBFS type (39,235 relabelled to `free_floating` by step S3, 1,630 to `carsharing` by step S1). We report two complementary uncertainty bounds on the 30.9 % removal rate to surface the role of clustering. A *station-level* Wilson 95 % interval, ignoring within-system correlation, gives $[30.54\,\%, 31.24\,\%]$; this is the bound a naive independent-Bernoulli assumption would place on the figure and is a *lower bound* on the real uncertainty. A *size-stratified cluster bootstrap* (123 systems partitioned into size quartiles, resampled with replacement within each stratum, $n = 10{,}000$, seed 42) does not yield an informative interval, because the largest size quartile contains a single dominant deployment (*Vélib' Métropole*, $\sim$ 14,700 stations) whose inclusion or exclusion across resamples mechanically shifts the denominator by an order of magnitude. The contrast between these two estimates, one tight and one degenerate, is itself the substantive observation: on a 123-system corpus with size spanning four orders of magnitude, the within-snapshot uncertainty on a stations-level total is dominated by which operators happen to be present, not by sampling noise within operators. We therefore report the 30.9 % removal rate as a deterministic accounting fact on the audit-dated snapshot, and defer temporal uncertainty to experiment E3 (twelve-month stability) where re-running the audit on freshly fetched snapshots gives a properly aligned variance estimate. The headline figure is that fewer than 9 % of raw entries survive into the catalogue without either being relabelled or removed, a strong empirical argument against the view that the standard alone delivers a research-ready dataset.

Table 2: Effect of the purging pipeline on the national GBFS corpus. *Certified* entries are kept in the Parquet with a typed `station_type` so that downstream consumers can filter explicitly; *rejected* entries are kept in a separate `rejected_stations.parquet` with the exclusion motive.

Indicator	**Raw**	**Audit Catalogue**
Inventoried GBFS systems	142	123 (certified)
Cities covered	–	97
with at least one dock-based station	–	64
Total stations (certified)	67,000	46,307
dock-based	–	5,442
free-floating (labelled, preserved)	–	39,235
car-sharing (A1)	1,630	re-labelled, excluded from BSS use
Stations dropped from catalogue	–	20,693 (30.9 %)

The most heavily impacted urban areas are those operated by free-floating fleets (*e.g.*Bordeaux, Nice, Saint-Étienne) and those historically lumped together with car-sharing systems on the national portal (Strasbourg, Grenoble), whereas urban areas served by traditional dock-based operators (Vélib' in Paris, VéloMagg in Montpellier) are left almost intact. After purging, the certified-station mass concentrates in Île-de-France (the historical Vélib' deployment), while regions with many free-floating systems (Nouvelle-Aquitaine, Auvergne-Rhône-Alpes) hold a much smaller dock-based stock once the A3 correction is applied.

3.2 Case study: Bordeaux and A3

Bordeaux is the paradigmatic case (Table 3). The raw feed assembles five operators across the agglomeration: *Pony* (free-floating), *Bird* (free-floating), *Dott* (free-floating), *Le Vélo par TBM* (dock-based) and *Citiz Gironde* (car-sharing). *Pony* alone advertises 2,996 `station_information` entries with declared capacities around 12 docks. Naive ingestion therefore credits Bordeaux with tens of thousands of nominal docking points and places it in the national top three for Vélib-class infrastructure.

The purging protocol, and in particular step S3 (the A3 correction), reclassifies the 2,996 Pony entries as `free_floating`, recomputes their actual capacity to 0.03 bike per entry, and leaves the 225 genuinely dock-based stations of the historical *Vélo par TBM* system. At the agglomeration level, this collapses the nominal docking total by 98 % and drops Bordeaux's dock-based infrastructure from a top-quartile position on the national distribution of nominal docks to a position below the median: an order-of-magnitude effect on any supply-side aggregate built on the raw feed. The twelve urban areas most exposed to A3 are mapped in Figure S3.

Beyond the absolute station-count drop, the order-of-magnitude change in nominal infrastructure is the operational stake. The five urban areas whose count of dock-based stations is most reduced by the purging protocol, together with the anomaly classes that drive the reclassification, are reported in Table S7. The shift is strikingly consistent: every one of these five urban areas would be credited with a top-decile dock-based deployment by a naive ingestion of GBFS, yet after the audit only Paris retains more than 1,500 certified dock-based stations. A non-audited

Table 3: Bordeaux: impact of the A3 correction on the *Pony Bordeaux* subsystem and on the agglomeration as a whole. The nominal docking total drops by 98.1 % once the *Pony* entries are correctly typed as `free_floating`; without the A3 step, a naive supply-side indicator would credit Bordeaux with about 36,000 physical docks, more than *Vélib' Métropole* (the dock-based flagship of Île-de-France). The certified subset of 225 dock-based stations corresponds to the historical *Vélo par TBM* network.

Indicator	**Raw GBFS**	**Audit Catalogue**
Pony Bordeaux subsystem		
`station_information` entries	2,996	2,996 (relabelled)
Declared as dock-based	2,996	0
Declared capacity per entry (median)	12	12
Recomputed actual capacity	–	0.03 bike/entry
Bordeaux agglomeration (all 5 systems)		
Total raw entries	9,921	–
Certified dock-based stations	–	225
Nominal docking total (dock-based only)	35,952	690
Implied over-count factor	–	$\times$ 52

pipeline does not just over-count by a small margin; it changes the relative positioning of the principal French agglomerations on any supply-side metric by an entire quartile.

Two further illustrative cases: Nice (A2) and Strasbourg (A1). A3 is not the only class with a measurable downstream effect. *Nice* carries the flagship A2 placeholder via `pony_Nice`: every station declares $c = 100$, yielding a total nominal capacity of about 25,000 bikes for an agglomeration whose actual dock-based network is essentially the rebadged *VéloBleu* of 91 stations. After purging, the nominal capacity collapses by a factor of 250 and Nice's national position on dock-count rankings shifts by an entire quartile. *Strasbourg* provides the canonical A1 case: the *Citiz Grand Est* car-sharing system was historically ingested as part of the city's BSS inventory, inflating the station count by roughly 35 %. After semantic exclusion the agglomeration loses that share of its nominal station count, on a part of the distribution where dock-count differences between French agglomerations are small and therefore more discriminating than at Bordeaux's scale. A1, A2 and A3 are therefore not interchangeable: each distorts a downstream metric through a different mechanism (false positive on the population, on the capacity, or on the type), and a non-audited pipeline conflates them.

3.3 Operator-level fragmentation of the capacity field

Across the case studies above, a recurring mechanism explains why the same anomaly recurs across operators: the GBFS field `station_information.capacity` is populated by French free-floating operators according to four mutually incompatible publication conventions, observed across six operator–city combinations (Table 4), aggregating to 84.7 % of the certified French rows. A downstream consumer who applies a single arithmetic to the field therefore aggregates four physically distinct quantities (NaN, constant placeholder, per-vehicle ratio, conditional fleet profile), silently mixed across six operator deployments. The corresponding operator-level pattern is global. Beryl London (UK) provides a positive control by publishing an empty `station_information` document for its free-floating fleet (zero entries), the canonical GBFS v3 behaviour for non-docked fleets, and is therefore not flagged by any of A1–A7. In particular, A7 only fires on *populated* documents whose entries declare `capacity = NaN`, which is the qualitatively different Dott pattern. The same operator's Greater Manchester deployment, by contrast, populates `station_information` with a 10.5 % zero-capacity rate (*cf.* Section 3.5) and is therefore A6-flagged: A6 and A7 thus carve out distinct publication regimes within a single operator's portfolio. The official MobilityData GBFS Validator [17], by contrast, returns no error on any of the flagged French systems (0 % recall on A3 in Table 5).

Table 4: Six observed semantics of the same `station_information.capacity` field across French free-floating operators. The six operator deployments enumerated below cover 39,004 stations; the residual 231 stations belong to smaller free-floating operators (Lime, Tier, Dott pilots, etc.) that follow one of the same four publication conventions and are folded into the corresponding bucket. Aggregate FF coverage is therefore 39,235 stations, 84.7 % of the certified corpus.

Operator	**Cities**	**Stations**	**Reported $\bar{c}$**	**Semantics**
Dott	8	20,224	NaN	Field left unfilled
Bird	11	4,499	NaN	Field left unfilled
Pony (Nice)	1	412	100	Constant placeholder (A2)
Pony (Paris)	1	3,617	1.6	Per-vehicle ratio
Pony (Bordeaux, Angers, ...)	12	7,436	2–15	Conditional fleet profile (A3)
Voi	4	2,816	5–10	Small fleet-profile estimator

3.4 Global scalability projection

Single-pipeline re-audit. To compare France and the world under *one* rule set rather than two, we re-fetched the MobilityData catalogue live and audited every reachable feed with the same function used on the French catalogue (`gbfs_toolkit.audit_static`, identical thresholds), archiving each raw feed with a UTC timestamp and SHA-256 so the audit is deterministic and seeded. Of the 1,509 inventoried systems, 936 publish a populated `station_information` and are audited (228,294 stations, 46 countries). The single-pipeline incidence with its cluster-bootstrap 95 % interval is A4 in 55.0 % [51.8, 58.2] of systems, A7 in 30.8 % [27.9, 33.7], A1 in 3.8 % [2.7, 5.1], A2 in 0.9 % [0.3, 1.5] and A5 in 3.2 % [2.1, 4.4] (Table 1, World column). Two results follow. First, A1 is feed-intrinsic, but in a different document: `station_information` carries no type, yet the GBFS v3 `vehicle_types` feed declares a `form_factor`, which 876 of the 936 systems publish. Reading car-sharing from `form_factor = car` flags 43 systems worldwide (16 French, matching the catalogue), and an operator-name heuristic is needed only for the $\sim$ 6 % of feeds lacking `vehicle_types`; A2, A4, A5 and A7 are intrinsic to `station_information` itself. Second, the feed-intrinsic anomalies recur at double-digit rates across 46 countries under thresholds calibrated on France, so the patterns are not French-specific. The per-country breakdown below is drawn from a frozen 2026-04 snapshot of the same catalogue, retained for its operator-level detail.

We applied the rule set, unchanged except for a country-perimeter recalibration, to the entire MobilityData canonical catalogue (1,509 GBFS systems across 48 countries, the complete world inventory at the audit date) as an exploratory projection of scalability beyond the French corpus. Because the semantic thresholds remain calibrated on French data without external ground truth, the figures below quantify the pipeline's reach and the gross incidence of each pattern rather than operator-level compliance in each country. Of these, 1,420 systems (94.1 %) were reachable, 917 publish a `station_information` document, and 204 systems trigger at least one A1–A5 class. The result rests on three methodological corrections that emerged from the audit itself. First, a country-perimeter calibration on A4. Second, a substring-matching fix in the A1 detector: the GBFS v3 form factor `cargo_bicycle` contains the string `car` and was initially flagged as A1, so the fix tightens A1 to exact list membership on the GBFS v3 form factor enum. Third, an explicit acknowledgement that the bounding-box surface estimator in A5 overestimates the operational area for elongated networks (Switzerland's national operators have a bbox area of 99,000 km^2 against a convex-hull area of $\approx$ 41,000 km^2): the shipped detector (`audit_pipeline.core._flag_a5`) retains the bbox formula for cross-corpus comparability, and the threshold 50,000 km^2 is calibrated on bbox; a convex-hull variant is documented as a refinement option in Section 5. The French national portal *transport.data.gouv.fr* indexes only 123 of the 255 French entries that the MobilityData catalogue lists, so the global sweep also exposes the incompleteness of the national regulatory pipeline. The corrected breakdown is visualised in Figure S5, with the full per-country counts in Table S6.

A3 (structural over-capacity) is detected on 29 non-French systems concentrated in Czech Republic (12) and Germany (9); A2 (placeholder capacity) on 34 non-French systems with Czech Republic again the hotspot (16); A1 (out-of-domain car-sharing labelled as BSS) on 29 non-French systems concentrated in the DACH region (Germany 21, Switzerland 8). Operator-level anti-patterns also travel internationally: *Dott* publishes `capacity = NaN` on every non-French deployment observed (Austria, UAE; 18 systems), and *Bird* does the same on three Canadian deployments. The cross-country audit also surfaces A6 (zero-capacity dock), the sixth class of the taxonomy, formalised in Section 2.3. The empirical conclusion is that the documented anti-patterns are not French-specific but global, and the protocol generalises to a 1,254-system corpus with no methodological recalibration.

Two national hotspots: monopoly (Czech Republic) versus fragmentation (Switzerland). The two most informative national hotspots are organisational opposites. In the **Czech Republic**, all 45 audited systems belong to a single operator (*nextbike*), and 25 trigger A2 (placeholder capacity, 16 systems) or A3 (over-capacity, 12), with 3 flagged by both: the exact cross-country mirror of *Pony*'s French free-floating A2/A3 hotspot. **Switzerland** is the dual configuration. It is the only country besides France to trigger all five structural classes (8 A1, 3 A2, 3 A3, 1 A4, 5 A5 across 16 of 49 systems), but through *fragmentation*: national car-sharing brands listed on the federal catalogue trigger A1[1]; mixed scooter/bike fleets (*Voi*, *Bolt*, *nextbike Switzerland*) publish capacity profiles that trigger A3; and federal multi-operator aggregators (*sharedmobility.ch*, *Velospot*) emit a single feed whose union bounding box mechanically exceeds the A5 macro-region threshold. The same A1–A5 envelope is thus reached through opposite configurations, monopoly publication (CZ) or aggregation-layer fragmentation (CH); the publisher-agnostic audit detects both and records the operator-versus-aggregator distinction so downstream consumers can disambiguate the A5 attribution.

Italy: the NaN-capacity blind spot that motivates A7. *Italy* is the boundary case that justifies A7: of the 25 systems with published station data, none is flagged by A1–A6, yet 17 are *Dott* deployments (Milan alone publishes 11,199 stations) declaring `capacity = NaN` on every entry. Because A2 needs a constant non-zero value, A3 needs numerical capacities, A4–A5 are geospatial and A6 tests $c = 0$ (not `NaN`), the pattern is invisible to A1–A6. A7 closes the gap as a consumer-side warning (Section 2.4): 245 systems with at least 20 stations cross the 50 % NaN-capacity threshold and are flagged exclusively by A7, covering 97,314 stations, about 76 % of them *Dott* alone (147 systems across Germany, Poland, Italy, Austria and beyond). The point is not non-compliance but non-aggregability: a consumer summing this column would silently mix `NaN` with real dock counts.

3.5 Incidence of the semantic warnings A6 and A7

The empirical distributions that calibrate the A6 and A7 thresholds (Sections 2.3 and 2.4) are reported in the Supplementary Materials (Tables S2 and S3); the resulting per-class incidence on the audited corpus is summarised here.

[1] An initial A1 run over-flagged 64 German systems because the detector substring-matched `"car"` inside the GBFS v3 `cargo_bicycle` form factor; tightening A1 to exact form-factor enum membership cut this to 21 genuine car-sharing systems and the global flag count from 215 to 173, a reminder that open audit pipelines need their own audit.

A6 incidence on the audited corpus. At $\tau_{A6} = 1\,\%$, 22 of the 301 docked systems are flagged. The extreme tail is Vélo Fluo Grand-Est (France, 38.5 %), Beryl Greater Manchester (UK, 10.5 %), MonaBike (Monaco, 6.8 %), Bicicoruña (Spain, 5.9 %), Vel'in (5.0 %), MBajk (Slovenia, 4.8 %), CyclOcity (Japan, 4.2 %) and àVélo (Canada, 3.5 %). On the certified French catalogue, no dock-based system crosses the threshold (0/65 dock systems with $c = 0$), a negative control consistent with the dock-based baseline.

A7 incidence on the audited corpus. On the 640 globally audited systems with at least 20 stations, 245 systems cross the A7 threshold and are simultaneously not flagged by any of A1–A6, covering 97,314 stations. The dominant operator is *Dott* (147 systems across Germany, Poland, Italy, Austria and beyond, 76 % of the exclusive-A7 station mass), followed by *Bird* (21 systems) and *nextbike* (15 systems where the operator chose NaN rather than placeholder), plus several smaller operators. The empirical conclusion is that the same publication choice that affects the French Dott and Bird fleets is propagated internationally, and that the v1.0 taxonomy A1–A5 captures only the visible failure modes of the field; A6 and A7 jointly close the principal blind spots that remain on the `capacity` channel.

4 Validation and robustness

4.1 Comparison with naive baselines

To check whether the full pipeline is justified or whether simpler heuristics suffice, we evaluate three baselines on the same 123-system corpus (Table 5). The *Raw GBFS* baseline ingests `station_information` as is. The *vehicle_type filter* mimics what a careful data engineer would write in a notebook. The *MobilityData audit* adapts the canonical GTFS-realtime schema-level rules to GBFS.

Table 5: Measured comparison of the Audit Catalogue against three baselines on the same 123-system French GBFS corpus. Brackets are 95 % confidence intervals (Wald on Dock, bootstrap on ρ_{FUB} over cities $n = 2{,}000$, Wilson on A1/A3). "Dock" for the Audit Catalogue is the reference by construction.

Pipeline	**Dock**	$\rho_{\textbf{FUB}}$	**A1 ex.**	**A3 fix.**
Raw GBFS	0.12 [.12, .12]	0.38 [.01, .68]	0 %	0 %
`vehicle_type` filter	0.15 [.15, .15]	0.19 [−.20, .52]	82 % [59, 94]	0 %
MobilityData audit	0.15 [.15, .15]	0.19 [−.20, .52]	82 % [59, 94]	0 %
Audit Catalogue	**1.00**	0.25 [−.13, .59]	**100 %** [82, 100]	**100 %** [91, 100]

A3 reclassification recall is 0 % for every baseline (Wilson 95 % upper bound $\approx 8.6\,\%$ at $n = 41$ A3-positive systems): structural over-capacity is invisible to syntactic validators by construction, and only a semantically-aware step on the capacity distribution can recover it. The *vehicle_type filter* and the *MobilityData audit* baselines produce identical numbers on this corpus, because schema-level checks catch nothing the name filter has not already excluded; French GBFS feeds are syntactically well-formed but semantically biased. Finally, ρ_{FUB} is highest on the raw count and drops after audit, which is expected rather than a regression: FUB captures perception of the entire cycling environment, so the raw count over-fits the perceptual signal by including virtual stations, while the Audit Catalogue's lower correlation reflects an honest count of physical docks.

4.2 Topology-aware geospatial detection (A4 v2)

The original A4 detector (step S5 in Algorithm 1) flagged every station whose haversine distance to the system centroid $\boldsymbol{\mu}_s$ exceeded $\sigma_{\max}$ robust standard deviations (MAD $\times 1.4826$). This isotropic assumption fails on two network geometries commonly observed in the global catalogue: linear networks ($r = \lambda_1/\lambda_2 > 5$ on the 2D coordinate covariance, *e.g.*riverfront corridors) and multi-hub national operators. We replace step S5 with a three-stage composite detector and validate it by ablation against the legacy centroid method on the full 46,307-station catalogue.

Stage 1 (density). HDBSCAN [38, 39] clusters stations by haversine distance with `min_cluster_size` $= \min(5, \lfloor n/3 \rfloor)$ and returns a per-station outlier probability $o_i \in [0, 1]$.

Stage 2 (spectral). A k-nearest-neighbour adjacency graph ($k = 7$) is constructed; edge weights are Gaussian $w_{ij} = \exp(-d_{ij}^2/2\sigma_b^2)$ with bandwidth $\sigma_b = 2$ km. The normalised graph Laplacian $\mathbf{L}_{\text{sym}} = \mathbf{I} - \mathbf{D}^{-1/2}\,\mathbf{W}\,\mathbf{D}^{-1/2}$ is eigendecomposed; the spectral isolation score s_i is the ℓ^2-norm of station i's embedding in eigenvectors $2 \ldots K$ ($K = 10$), normalised to $[0, 1]$. A station that sits in a separate spectral component (near-zero Fiedler value λ_2) is spectrally disconnected from the network core. We adopt the symmetric normalised Laplacian $\mathbf{L}_{\text{sym}}$ following von Luxburg [40]; an ablation against the random-walk Laplacian $\mathbf{L}_{\text{rw}}$ on the full catalogue yielded $< 0.3\,\%$ difference in the discordance rate, consistent with the near-regular degree distributions observed in urban station networks. The eigendecomposition uses a sparse eigensolver (`scipy.sparse.linalg.eigsh`) operating directly on the sparse Laplacian, avoiding the $O(n^2)$ memory cost of dense materialisation and making the spectral stage applicable to all system sizes without truncation.

Stage 3 (fusion). The composite score $a_i = \alpha\,\hat{o}_i + (1 - \alpha)\,s_i$ with $\alpha = 0.5$ fuses density and spectral isolation. A station is flagged A4 when $a_i > 0.8$. A sensitivity analysis over $\alpha \in \{0, 0.25, 0.5, 0.75, 1.0\}$ and threshold $\in \{0.6, 0.7, 0.8, 0.9\}$ shows that the Top-10 most-flagged systems are invariant for $\alpha \geq 0.25$ and threshold ≥ 0.7; the default ($\alpha = 0.5$, threshold $= 0.8$) sits in the stable region. To classify each system's spatial geometry we compute the eigenvalue ratio $r = \lambda_1/\lambda_2$ of the 2D coordinate covariance matrix: $r > 5$ indicates a *linear* network.

Table 6 summarises the ablation (visual overview in Figure S1). The composite detector agrees with the legacy method on 80.6 % of stations. Of the discordant stations, 17.3 % are flagged by the legacy method only (centroid, not composite), while only 2.2 % are new findings of the composite detector. Three cases illustrate the gain: *Paris* (1,507 stations) drops from 260 legacy flags to 43 composite flags (saving 217 false positives); *Dott-Paris* (14,747 stations) drops from 2,750 to 0; and *Clermont-Ferrand* (82 stations, linear network with $r = 6.07$) drops from 18 to 0. The pre-registered human audit of Section 4.5 adjudicates both sides of this disagreement: it confirms the removed legacy flags as false positives (14/14) but does not confirm the bulk of the composite-only candidates, which the production fusion threshold of 0.8 already discards.

Table 6: Ablation of the legacy centroid-based A4 detector (3σ MAD) against the topology-aware composite detector (HDBSCAN + spectral graph) on the 46,307-station catalogue. AGREE = both methods concur; DISCORDANT_LEGACY = flagged by centroid only; DISCORDANT_COMPOSITE = flagged by composite only.

Discordance class	**Stations**	**%**
AGREE_CLEAN (neither flags)	36,917	79.7
AGREE_FLAG (both flag)	384	0.8
DISCORDANT_LEGACY (centroid only)	8,005	17.3
DISCORDANT_COMPOSITE (composite only)	1,001	2.2
Total	46,307	100.0

4.3 Leave-one-operator-out cross-validation

To check that the rules are not tuned to a handful of operators (*e.g.*Pony, Dott, Bird), we evaluate their cross-operator generalisability with a *leave-one-operator-out* (LOOO) cross-validation: for each of the K eligible operators (≥ 50 stations, ≥ 2 systems), the rule thresholds are estimated on the remaining $K-1$ operators and applied, unchanged, to the held-out operator. The per-rule flag rate on the test fold is recorded; inter-fold stability is measured by the coefficient of variation $\mathrm{CV} = \sigma/\mu$ of the flag rate across folds. A $\mathrm{CV} < 0.20$ indicates that the rule is operator-agnostic. Seven operators are eligible: Bird, Citiz, Dott, Pony, Voi, Vélib', Vélo&Co. The low K limits statistical power; the LOOO should therefore be interpreted as a consistency check on rule stability rather than a definitive generalisability proof. Extension to the global catalogue ($K \approx 20$–30) is planned as follow-up.

Table 7 reports the results, with the overview in Figure S2. A crucial distinction emerges between *structural rules* and *threshold rules*:

- **A1** and **A3** are deterministic functions of `station_type`. Their high CV (2.45 and 0.87) reflects the fact that car-sharing operators trigger A1 at 100 % and free-floating operators trigger A3 at 100 %, while dock-based operators trigger neither. This is not overfitting; it is the definitional semantics of the rule.
- **A7** ($\mathrm{CV} = 1.19$) identifies a genuine industrial anti-pattern: Bird (100 %) and Dott (99.96 %) systematically publish `capacity = NaN`, while the remaining operators do not. The LOOO result proves that the A7 signal is operator-specific *and real*, not an artefact of threshold tuning.
- **A6** ($\mathrm{CV} = 0$) passes trivially: no operator triggers zero-capacity dock in this corpus.
- **A4** ($\mathrm{CV} = 1.02$, flag rate 4.7 % $[1.7, 8.7]$ 95 % bootstrap CI) shows moderate operator dependence, consistent with the geometry-specific sensitivity documented in Section 4.2.

The LOOO also validates hypothesis H3 (the rules do not fire on clean operators): holding out Vélib' Métropole (1,533 dock-based stations), the test-fold flag rate is 0 % on A1–A3, A5–A7 and only 1.7 % on A4, consistent with residual GPS noise. Holding out Vélo&Co (94 stations) yields 0 % across all seven rules.

Table 7: Leave-one-operator-out cross-validation on the 46,307-station French catalogue. $K = 7$ eligible operators. CV = coefficient of variation of the per-fold flag rate; CI = 95 % bootstrap confidence interval on the mean flag rate, resampled over stations within each fold ($n = 500$ iterations).

Rule	**CV**	**Mean flag rate**	**95 % CI**	**Interpretation**
A1	2.45	14.3 %	[0, 43] %	Structural (type-based)
A2	2.45	0.5 %	[0, 1.5] %	Rare; one operator only
A3	0.87	57.1 %	[14, 86] %	Structural (type-based)
A4	1.02	4.7 %	[1.7, 8.7] %	Geometry-dependent
A5	2.45	3.0 %	[0, 9.1] %	National-scale operator
A6	0.00	0.0 %	[0, 0] %	No trigger in corpus
A7	1.19	37.8 %	[9.3, 71.4] %	Real anti-pattern

4.4 Internal robustness checks

Two algorithmic checks complement the human validation of Section 4.5, and a third tests threshold sensitivity.

Orthogonal coverage. Three rule-based classifiers, each restricted to one signal source (name, capacity, geospatial), agree on a stratified 500-station sample (seed `numpy.random.default_rng(42)`, shipped with the audit report) at essentially chance level (Fleiss $\kappa = -0.07$): the three sources are not redundant. Their *union* nonetheless reaches 97.2 % recall at 91.5 % precision against the integrated pipeline, with only 7 of 500 stations flagged by all three, so A1–A5 cannot be reduced to a single heuristic.

Full-task agreement. Three decision-tree classifiers seeing all signals but in different priority orders reach Fleiss $\kappa = 0.80$ on the same sample (pairwise $\kappa \in [0.70, 1.00]$), with disagreements concentrated on the A1/A3 frontier.

Threshold sensitivity. A sensitivity sweep of the A4 outlier detector over $\sigma \in \{2.0, 2.5, 3.0, 3.5, 4.0\}$ leaves the downstream count-level summaries essentially invariant: Kendall's τ between the $\sigma = 3$ reference and each alternative stays in $[0.91, 1.00]$ on both the Top-10 cities and the Top-10 operators by certified docked-bike station count, and Top-10 city membership is invariant across the grid. The $\sigma = 3$ choice therefore sits on a plateau of stable downstream classification. The full sweep and a methodological caveat on the MAD scale estimator are reported in the Supplementary Materials (Table S1).

Multi-threshold robustness. The single-dimension pilot above generalises to a reproducible sweep of every policy threshold, exposed as parameters of the released audit library and driven by its `audit_sensitivity` routine. Sweeping the A4 robust-sigma ($\sigma \in [2, 4]$), the A5 bounding-box area ($[30{,}000, 70{,}000]\,\mathrm{km}^2$), the A6 and A7 rate thresholds ($\tau_{A6} \in [0.005, 0.05]$, $\tau_{A7} \in [0.3, 0.7]$) and the minimum system size ($N_{\min} \in [10, 30]$), and measuring the Jaccard overlap of each flagged-system set against the published baseline, the flagged set is *invariant* (Jaccard $= 1.00$) for A1, A2, A3 and A6 across every grid, and stays on a stability plateau for the geospatial and capacity rules (A4 Jaccard ≥ 0.91 over the σ grid, A7 ≥ 0.97 over the τ_{A7} grid). The published verdicts therefore do not hinge on the exact threshold values. Each per-class incidence we report additionally carries a cluster-bootstrap 95 % confidence interval (systems resampled with replacement, seed pinned) produced by the same library; the French A3, A4 and A7 incidences are 0.33 $[0.25, 0.41]$, 0.72 $[0.63, 0.80]$ and 0.26 $[0.19, 0.34]$ of the certified systems. The sweep and the intervals are regenerated by a single call, so the robustness claim is itself reproducible rather than a one-off pilot.

These algorithmic checks complement, but do not replace, the human construct-validity check of Section 4.5; they are reported as robustness measures, not as substitutes.

4.5 Construct-validity check via blind, pre-registered author annotation

The ablation of Section 4.2 and the cross-validation above compare *algorithms*; neither can establish which verdict matches the physical world. We therefore re-examined a stratified sample of $n = 422$ catalogue stations against external imagery in a *pre-registered* campaign: frozen sample (with its SHA-256), locked codebook and analysis script, all fixed before any label was recorded (Section 4.6). The two authors labelled every station independently, blind to the pipeline output and to each other, and every number below uses concordant judgements only, so the estimates are conservative by construction. This is an internal construct-validity check whose credibility rests on pre-registration rather than on annotator independence (Section 5.5).

The campaign delivers three results. **(i)** On the decisive discordant-legacy stratum, *every* station with a concordant proximity judgement was judged connected to its network: the legacy-only A4 flags are confirmed false positives at 100 % (14/14, Wilson 95 % CI $[78.5, 100]$ %), with no confirmed true positive anywhere in the stratum. **(ii)** The redesigned instrument measures the judgements that carry this conclusion with near-ceiling reliability: the network-position question that collapsed at $\kappa = 0.13$ in campaign v1 reaches AC1 $= 0.946$, and the perimeter question 0.983. **(iii)** At catalogue level, the audit's dominant flag is overwhelmingly justified (96.8 % of adjudicated A3 flags mark genuinely defective records), and the campaign quantifies a defect class previously invisible to feed-only audits: one adjudicated random station in five is a *ghost*, with no service infrastructure on the ground.

Inter-rater reliability. Table 8 reports agreement per decision-tree node. The two geospatial nodes exceed the pre-registered target (AC1 ≥ 0.70) by a wide margin, 0.983 for the perimeter question and 0.946 for network proximity against $\kappa = 0.13$ for the equivalent v1 question: the one-question-at-a-time redesign repaired exactly the judgements that carry the audit's geospatial verdicts. The table also shows why AC1 was pre-registered as the primary coefficient: on Q3a both annotators judge ~99 % of stations in-perimeter, so Krippendorff's α collapses to -0.006 *despite* 98 % raw agreement, the textbook prevalence paradox [41].

Table 8: Inter-rater reliability per decision-tree node, on co-routed stations, indeterminate pairs excluded from the coefficients and reported as a rate. The pre-registered primary coefficient is Gwet's AC1 [41], robust to the prevalence paradox that deflates Krippendorff's α [42] and Cohen's κ [43] on skewed marginals (both are shown, with the modal-class share, so the paradox stays visible). Pre-registered success target: AC1 ≥ 0.70; a node below 0.60 is flagged as an instrument weakness. Q2 elicits a raw count, for which nominal coefficients are not meaningful; its agreement is reported in the text.

Node	**AC1**	P_o	**Modal**	α	κ	**Indet.**	n
Q0 – infrastructure type	0.559	63 %	49 %	0.449	0.456	7.9 %	386/419
Q2 – dock count		numeric node; see text				6.3 %	30/32
Q3a – perimeter	**0.983**	98 %	99 %	−0.006	−0.006	0.8 %	237/239
Q3b – network proximity	**0.946**	95 %	96 %	0.309	0.319	0.0 %	239/239
Composite verdict (secondary)	0.532	56 %	46 %	0.375	0.382	–	422

The two remaining nodes localise the residual difficulty without touching those verdicts, and their disagreement patterns are findings in their own right. The infrastructure-type node Q0 sits below the pre-registered target (AC1 = 0.559, reported as an instrument weakness per protocol), and its divergences are systematic rather than random: scooters-only versus free-floating bikes on mixed-fleet operators (26 cases in one direction, none in the other), an empty car-sharing bay versus nothing visible (24 cases), a virtual free-floating zone versus nothing visible (49 cases). All three are boundary objects that offer no observable hardware, precisely the cases where the GBFS specification is silent on what a "station" physically is: the node fails where the standard itself underspecifies, which is the gap the taxonomy formalises and the operator recommendations of Section 5.5 address. The dock-count node Q2 makes the same point for capacity: even two domain experts on shared imagery cannot reliably elicit a dock count (exact agreement 33 %, $n = 30$), independent corroboration that declared capacity is not auditable without ground truth. Rule A2 accordingly keeps its system-level parser-fidelity validation (Table 9).

Per-rule performance. Table 9 confirms the flags that dominate the catalogue. A3, which carries the audit's headline reclassification, reaches recall 0.99 [0.93, 1.00] and marks a genuinely defective record in 96.8 % [93.2, 98.5] of adjudicated cases (183/189): only 6 of 189 flags landed on healthy infrastructure. Its class-exact precision (0.73 [0.49, 0.82]) is lower for an instructive reason: 45 of the 51 nominal false positives are stations where the annotators found *nothing at all*, defects more severe than the flag asserts rather than healthy stations wrongly flagged. A1 reaches recall 0.91 with precision 0.71, and three of its four false positives are empty car-sharing bays judged "nothing visible" on the imagery, the boundary object above, so they indict the imagery at least as much as the rule. A4's per-station sample is small but clean (2 TP, 0 FP); its decisive test is the ablation below. A5 fires on declared-perimeter inconsistencies that are invisible in imagery (none of its 12 assessable flags was visibly misplaced), so its construct, like A2's and A7's, is validated at the system level instead (Section 4.3). The triage framing of these rules is unchanged: A3–A5 are high-recall screening filters whose false positives cost one human glance each, and the decisive question, whether the flags the topology-aware detector *removes* are genuine false positives, is answered next.

The A4 ablation, adjudicated. The campaign's decisive result adjudicates the disagreement of Table 6 between the legacy centroid detector and the topology-aware composite. On the *discordant-legacy* stratum (sampled from the 8,005 stations the legacy method alone flags), all 14 stations with a concordant proximity judgement were judged connected to their network: a legacy false-positive rate of 100 % (Wilson 95 % CI [78.5, 100] %), with no confirmed true positive

Table 9: Per-rule performance against the adjudicated reference, derived *a posteriori* from concordant node answers (never from a pipeline-referencing verdict), pooling the SRS and the booster strata. Intervals: cluster-robust 95 % bootstrap over systems (2,000 resamples, seed 42); A4, with too few positives for a stable bootstrap, reports the Wilson interval [44]. A5 recall is not estimable (no adjudicated out-of-perimeter station).

Rule	Real iff	TP	FP	FN	Precision [95 % CI]	Recall [95 % CI]
A1	car-sharing (Q0)	10	4	1	0.71 [0.42, 0.93]	0.91 [0.67, 1.00]
A3	no physical dock (Q0)	138	51	2	0.73 [0.49, 0.82]	0.99 [0.93, 1.00]
A4	disconnected (Q3b = no)	2	0	1	1.00 [0.34, 1.00]	0.67 [0.21, 0.94]
A5	out of perimeter (Q3a = no)	0	12	0	0.00 [0.00, 0.24]	–
A2	not per-station assessable (no concordant dock count); system-level parser fidelity only					
A6	no eligible station in the sample (no docked station with declared capacity 0)					
A7	system-level semantic warning, not imagery-validatable (252 flagged stations, 30 systems)					

anywhere in the 40. This replicates, under a stricter pre-registered instrument, campaign v1's 46/46 on the same population; the two campaigns together leave the conclusion in little doubt. The *discordant-composite* stratum completes the picture: 9 of its 21 adjudicated stations are ghosts (nothing on the ground, a defensible flag) and one of the 12 with a concordant proximity judgement is confirmed out-of-network. The production rule anticipates exactly this asymmetry: at the shipped fusion threshold of 0.8 (step S5, Section 2.5) none of the 40 sampled composite-only stations is flagged, so the certified catalogue removes the 8,005 legacy false positives *without* inheriting speculative new flags. The human check thus validates both halves of the design choice: what the composite detector removes is wrong, and what it declines to add is not missed.

Catalogue-level prevalence. Table 10 estimates how common each class is, on the random SRS only ($n = 300$; the booster strata over-represent rare classes by design). The flag rates are deterministic properties of the audit-dated snapshot; the confirmation column is conditional on concordant labels (190 of 300 stations are decidable at Q0). Two findings stand out. First, the dominant A3 flag (89.3 % of the SRS) is overwhelmingly justified: 76.3 % of flagged, decidable stations are confirmed free-floating points presented as stations, rising to 96.5 % [92.6, 98.4] once ghost stations are counted as the defects they are. Second, the campaign delivers what is, to our knowledge, the first quantified ghost-station rate for a national GBFS corpus: 37 of the 300 sampled stations (12.3 %, Wilson [9.1, 16.5] %, a lower bound that counts every non-adjudicated station as healthy; 19.5 % [14.5, 25.7] among decidable stations) show no service infrastructure at the declared coordinate. The audit already surfaces almost all of them, 35 of the 37 carrying an A3 flag, so the certified catalogue quarantines these records today; promoting them to a dedicated ghost-station class requires the dynamic `station_status` evidence deferred to the L2 audit layer and is the clearest item on the roadmap (Section 5.5).

4.6 Human annotation protocol (pre-registered)

Campaign v1 asked four parallel factual questions of every station and two of them collapsed (capacity coherence at $\kappa = 0.17$, network position at $\kappa = 0.13$): interpretive phrasing, asked off-context. Campaign v2 replaces the instrument with a *one-question-at-a-time decision tree* and pre-registers the entire chain: frozen 422-station sample with its SHA-256, locked codebook with worked examples, and the analysis script fixed before any label was recorded; post-freeze changes are barred and, where unavoidable, disclosed as such.

Instrument. The first node Q0 classifies the *physical infrastructure* (not the transient presence of vehicles) into five types: docks with fixed anchor hardware, a ground free-floating zone, scooters only, a car-sharing bay, or nothing visible, plus an explicit *indeterminate* escape at every node.

Table 10: Catalogue-level prevalence of each class on the random prevalence sub-sample only ($n = 300$ SRS), with Wilson 95 % intervals; booster strata excluded by design. *Flag rate*: pipeline flag on the audit-dated snapshot. *Human-confirmed*: adjudicated confirmation among flagged SRS stations with concordant labels.

Class	**Flag rate (SRS)**	**Wilson 95 % CI**	**Human-confirmed**
A1 (car-sharing)	2.0 %	[0.9, 4.3]	3/3 flagged confirmed
A2 (placeholder)	0.0 %	[0.0, 1.3]	no flag on the SRS
A3 (free-floating)	89.3 %	[85.3, 92.3]	132/173 class-exact; 167/173 defective
A4 (geospatial)	1.0 %	[0.3, 2.9]	2/3 ghost at the point; 0 assessable for Q3b
A5 (out-of-perimeter)	1.7 %	[0.7, 3.8]	0/3 (see construct mismatch above)
A6 (zero-capacity dock)	0.0 %	[0.0, 1.3]	no flag on the SRS
A7 (null capacity)	73.0 %	[67.7, 77.7]	system-level, not imagery-validatable

A docked station receives a raw dock count (Q2); any located object receives two placement questions, geographic perimeter (Q3a) and network proximity (≤ 1 km from a sibling or inside the network hull, Q3b). Per-rule ground truth is derived *a posteriori* from the adjudicated answers, never from a pipeline-referencing verdict: A1 iff car-sharing; A3 iff no physical dock; A4 iff Q3b = no; A5 iff Q3a = no; A6 iff docked with declared capacity 0; A2 iff the counted docks are inconsistent with the declared capacity (± 50 %); A7 is evaluated at the system level only. The annotators (an urban-mobility researcher and a data engineer) judge from Street View, satellite imagery and OpenStreetMap with a sibling-station overlay; no stratum label or pipeline output is ever displayed.

Sampling ($n = 422$, seed 42). The design separates two goals, because the publication variance lives at the operator level. (i) A simple random sample of 300 stations from the 46,307-station catalogue estimates catalogue-level prevalence (± 5.7 % at 95 %) and validates the abundant classes. (ii) Operator-balanced booster strata (round-robin across distinct systems) cover the rare or decisive classes: the two A4 discordant sets ($40 + 40$, the ablation of Section 4.2), A1 (30 stations spanning all 17 French car-sharing systems), A5 (8) and A2 (4). Car-sharing is excluded from the geospatial strata (it terminates at Q0). A2 and A5 are mono- or pauci-operator in the French corpus and validate parser fidelity, not rule universality.

Adjudication and analysis. Disagreements are resolved by deterministically re-applying the locked codebook to re-fetched imagery; genuinely ambiguous cases are marked indeterminate and excluded, never defaulted for or against the pipeline, and the raw inter-rater confusion matrices are released so a reviewer can recompute alternative bounds. Because several strata have extreme class prevalence, the pre-registered primary reliability coefficient is Gwet's AC1 [41], reported with raw agreement, the modal-class share, Krippendorff's α and Cohen's κ so the prevalence paradox stays visible. Per-rule precision and recall carry a cluster-robust 95 % interval bootstrapped over systems (2,000 resamples, seed 42) alongside the Wilson interval, since stations within an operator are not independent. Street-View-staleness sensitivity, an attention-drift test on per-station timing and an intra-rater test-retest round are pre-specified. The frozen sample, codebook, annotation tool and analysis script are released (`experiments/human_validation/`).

4.7 Cross-country transferability pilot

To verify out-of-sample transferability of A1–A7 beyond French publication practice, we constructed a panel of thirteen non-French systems audited *live* via their public GBFS feeds, with the *same* rule set used on the French corpus and only the national perimeter recalibrated. All other thresholds ($\sigma = 3$, $N_{\min} = 20$, A2 placeholder rule, A3 capacity-profile ratio threshold of 5.0, $\tau_{A6} = 0.01$, $\tau_{A7} = 0.5$) are kept at their French values. The panel diversifies on the station-management *technology stack*, the layer at which publication conventions actually originate, rather than only on country: six distinct platforms are represented (PBSC, JCDecaux Cyclocity, Smartbike Mobility Plus, De Lijn / Blue-bike, Urban Sharing AS, Deutsche Bahn), spanning Spain, Belgium, Italy, Norway and Germany. Of the thirteen, twelve are metropolitan dock-based deployments and serve as negative controls; the thirteenth, *Call a Bike* (Deutsche Bahn), is the German nationwide system and is reported as an informative case rather than a control (its A5 / A7 firings are honest at national scope and its high A4 rate surfaces a known limitation of the single-centroid detector on multi-modal geometries). Results are reported in Table 11; the live-audit script (`scripts/audit_live_systems.py`) and the per-system machine-readable results (`experiments/e5_europe/results.csv`) ship with the source repository, so the audit can be replayed against the same set of public feeds on demand.

Table 11: E5 panel: A1–A7 rule set applied *live* to thirteen non-French systems on 2026-05-13 (`scripts/audit_live_systems.py`), with French thresholds and country-appropriate perimeter only. Six station-management platforms (PBSC, JCDecaux Cyclocity, Smartbike, De Lijn / Blue-bike, Urban Sharing AS, Deutsche Bahn) across five countries; twelve metropolitan dock-based deployments form the *negative-control panel*, plus one *informative case* (*Call a Bike Deutsche Bahn*, nationwide). **No metropolitan system triggers A1, A2, A3, A6 or A7**; A4 fires only on isolated stations ($\leq 4.5\,\%$) and A5 once on Sevici (sentinel-coordinate artefact, footnote *).

System	Country	Stack	Stns.	A1	A2	A3	A4 (%)	A5	A6	A7
Metropolitan negative-control panel										
Bicing Barcelona	ES	PBSC	545	—	—	—	0.18 %	—	—	—
bicimad Madrid	ES	PBSC	637	—	—	—	0.00 %	—	—	—
Bilbao Bizi	ES	PBSC	48	—	—	—	0.00 %	—	—	—
Bizi Zaragoza	ES	PBSC	276	—	—	—	0.36 %	—	—	—
Dbizi Donostia	ES	PBSC	70	—	—	—	2.86 %	—	—	—
Sevici Seville	ES	Cyclocity	247	—	—	—	4.45 %	✓*	—	—
Valenbisi Valencia	ES	Cyclocity	279	—	—	—	0.72 %	—	—	—
Velo Antwerpen	BE	Smartbike	321	—	—	—	0.00 %	—	—	—
Blue-bike	BE	De Lijn	276	—	—	—	0.72 %	—	—	—
BikeMi Milan	IT	Urban Sh.	320	—	—	—	1.56 %	—	—	—
Oslo Bysykkel	NO	Urban Sh.	266	—	—	—	1.50 %	—	—	—
Bergen City Bike	NO	Urban Sh.	94	—	—	—	1.06 %	—	—	—
Informative case (nationwide system, not a metropolitan control)										
Call a Bike (DB)‡	DE	DB	1,601	—	—	—	1.8 %	✓	—	✓
French corpus (ref.)	FR	—	46,307	17 sys.	1 sys.	41 sys.	88 sys.	4 sys.	0 sys.	32 sys.

* Sevici's A5 firing is a coupling artefact: two stations carry sentinel coordinates $(0,0)$ that inflate the bounding box from $\sim 80\,\text{km}^2$ to $\sim 4.5\times10^6\,\text{km}^2$; the coupling is discussed in Section 5. ‡ *Call a Bike* (Deutsche Bahn) is a nationwide deployment, not a metropolitan control. Its three flags are honest at national scope (A7 by Dott-pattern, A5 by $320{,}000\,\text{km}^2$ bbox, A4 on 1.8 % of stations via the nearest-neighbour rule of Section 2.5) and confirm the design choice of an NN-based A4 over a centroid-based variant (which over-flagged at 37.8 % during development).

The capacity-profile ratio that defines the A3 signature on free-floating French fleets is essentially unity on every metropolitan dock-based system in the panel, across six distinct technology stacks. This is the out-of-sample confirmation called for in Section 5.2: the A3 detector does not over-fire on clean docks. Vehicle-type metadata on all twelve metropolitan feeds is populated cleanly, so A1 produces no false positive either. A4 (geospatial outlier) fires only on isolated stations ($\leq 4.5\,\%$ of any system's mass), in line with what would be expected from any GPS-anchored fleet; the single A5 firing (Sevici) is shown by Table 11 footnote * to be a deterministic consequence of two sentinel-coordinate stations rather than a genuine $> 50{,}000\,\text{km}^2$ deployment. Finally, the informative *Call a Bike* case shows that when a system genuinely extends to national scope, A5 and A7 fire *by design* (the deployment crosses the macro-region threshold; the capacity column is structurally NaN at national scope) and the high A4 rate reflects a known limitation of the single-centroid detector on multi-modal geometries, motivating a sub-system-clustering refinement in future work.

Empirical justification of $\tau_{A3} = 5$. The choice of the A3 threshold can be inspected on two complementary corpora reported jointly in Table S8: the released French parquet (the derivation set; Section 5.2 caveat applies) and the 267 systems with computable ratios from a live global sweep on 2026-05-13 across 33 countries (`scripts/global_a3_ratio.py`; results in `experiments/e2_threshold_sensitivity/global_a3_ratio.csv`). On the French corpus the distribution is sharply bimodal: the 81 dock-based systems sit in $[1.00, 1.93]$ (median 1.00, 95th percentile 1.04), the 5 free-floating systems exhibiting conditional-averaging publication sit in $[51, 514]$, and the band $[2, 50]$ is empty. On the global corpus the picture is broadly consistent but less clean: the [1.00, 1.10] bucket still dominates (73.8 % of the 267 systems), the top ten systems by ratio are exclusively free-floating operators (*Pony*, *nextbike*, *Voi*, *Bolt*) confirming that very high ratios track the conditional-averaging pattern, but the intermediate band $[2, 5]$ is no longer empty (15 systems, 5.6 % of the audited set). We retain $\tau_{A3} = 5$ as a conservative threshold that captures the clear-bias regime (the 31 systems globally with ratio ≥ 5, including all 19 systems with ratio ≥ 20) while explicitly acknowledging that 15 systems in the moderate band $[2, 5]$ are not flagged by the current rule.

A non-parametric kernel-density cross-check on $\log_{10}(\bar{c}_{\text{profile}}/\bar{c}_{\text{actual}})$ (the 98 systems with ratio > 1.01, after removing the spike at unity) places $\tau_{A3} = 5$ inside the density valley that separates the near-unity mode from the high-bias mode, confirming the threshold sits in a low-density separator the data itself identifies. The valley-finding curves, the soft/hard two-tier variant ($\tau_{A3}^{\text{soft}} \approx 3$, $\tau_{A3}^{\text{hard}} \approx 15$) and the code are provided in the supplementary material (`scripts/a3_kde_threshold.py`).

4.8 Retrospective hold-out validation

This subsection tests *construct validity*: do the rules generalise to systems that were not specifically inspected at calibration time? External validity (whether the rules generalise to systems that did not yet exist at rule-freeze) is the object of the prospective E1 execution and is deferred. Concretely, we executed a retrospective version of the pre-registered E1 protocol (`experiments/e1_holdout/PROTOCOL.md`) on two calendar windows of the Mobility-Data canonical catalogue. The shorter window is the catalogue diff between commit `15e34e6` (2025-11-11) and the rule-freeze commit (2026-05-13): 345 systems added in six months, 39 with non-empty `station_information` that could be audited. The longer window extends back to commit `4665739` (2025-04-29, twelve months): 481 systems added, 98 audited. The remainder are free-floating systems without `station_information`, below $N_{\min}$, or authenticated. The frozen rule set was applied unchanged on both windows; per-system results in `experiments/e1_holdout/held_out_analysis.md` (6-month) and `held_out_results_12mo.csv` (12-month).

H1 (rule-firing rate within ±30 % of the global derivation rate of 13.5 %, i.e., 204/1,509 systems). On the twelve-month window the A1–A5 firing rate is 11/98 = 11.2 % (Wilson 95 % CI [6.4 %, 19.0 %]), which falls inside the pre-registered acceptance band [9.5 %, 17.5 %] at the point estimate. The six-month window gives 7/39 = 17.9 % (Wilson [9.0 %, 32.6 %]), which exceeds the upper bound by 0.4 percentage points at the point estimate but whose Wilson interval still contains both the derivation rate and the entire acceptance band. Reading the two windows jointly, H1 passes: the rule-firing rate on out-of-sample systems is statistically consistent with the derivation-set rate, and the more conservative twelve-month estimate is in the band on both point estimate and confidence interval.

H2 (operator-driven A3/A7). On the twelve-month window, 84.2 % (48/57) of the systems flagged on A3 or A7 belong to the free-floating operator family identified on the derivation set (Dott, nextbike, Pony, Bird, Voi, Bolt, Lime, Tier, Donkey, Spin); the six-month window gives 81.2 % (13/16). Both estimates exceed the pre-registered acceptance threshold of ≥ 50 % by more than thirty percentage points, so H2 passes cleanly.

H3 (clean-operator invariance). H3 is not testable on either window. Neither the six-month nor the twelve-month held-out panel contains a single system using one of the six clean-operator platforms of Section 3.4 (PBSC, Cyclocity, Smartbike, De Lijn, Urban Sharing, Deutsche Bahn). H3 therefore awaits the prospective re-execution of the protocol against a future MobilityData snapshot that includes at least one clean-platform addition.

H1 and H2 pass on the twelve-month window. H3 is deferred to the prospective re-execution. The implications of these outcomes for the construct validity of the taxonomy are discussed in Section 5.2.

5 Discussion

5.1 Implications

The audit is intended as a scientific validation tier between GBFS publication and academic use, addressing the gap between regulatory publication and research-grade reuse that the open-government-data literature identifies as a recurrent adoption barrier [45]. Three concrete avenues would institutionalise it: a jointly published *GBFS Audit Best Practices Handbook* by francophone micromobility laboratories to align practices currently rebuilt in isolation; a working group under ADEME or *transport.data.gouv.fr* to formalise an extended GBFS schema with a certification framework; and a continuous validation chain running the pipeline monthly on the national feeds, with automated publication of the audit report and operator-side notification on regression. As the infrastructure-and-classification literature observes [46, 47], the most durable infrastructures are those whose governance is explicit and tied to an identifiable community of practice; moving the artefact from author-led to community-led governance is therefore an institutional challenge as much as a scientific one. The technical lesson of the audit, that a single field (`capacity`) cannot serve six different operational semantics, is also the empirical case for what Stonebraker and Çetintemel [48] call the end of "one size fits all" data architectures: open standards that attempt to span dock-based, free-floating and car-sharing fleets in the same schema accumulate semantic ambiguity exactly because they refuse to specialise.

5.2 Scope of the contribution

The taxonomy's primary discrimination axis is dock-based versus free-floating publication. Of the seven classes, A2, A3, A6 and A7 all surfaced from operator publication conventions specific to free-floating fleets (conditional-averaging profiles, NaN-capacity fields, zero-capacity ghost-docks); A1, A4 and A5 are fleet-type-agnostic but the largest country-level hotspots (Table S6) are themselves operator-driven (*Pony* and *nextbike* on A2/A3, *Dott* on A7). The twelve metropolitan clean-platform negative controls of Table 11 demonstrate that the rule set does not over-fire on dock-based deployments; the free-floating-native extension is the object of experiment E6, planned for 2027-Q2.

The taxonomy is inductive and still requires out-of-sample statistical validation. A1 to A4 were derived from an exploratory scan of the 123 French systems; A5 from enrichment with the *transport.data.gouv.fr* catalogue; A6 and A7 from the empirical distributions surfaced by the global 1,509-system sweep (Section 2, Provenance of the taxonomy). Detection rules were calibrated on the same data they are reported to flag, an inductive workflow whose construct-validity risk we address with two out-of-sample tests reported in the Results: the negative-control cross-country panel (Section 3.4, Table 11, twelve metropolitan clean-platform systems audited live) and the retrospective hold-out validation (Section 4.8, 98 audited systems added to the MobilityData catalogue in the twelve months preceding rule-freeze). H1 and H2 pass on the twelve-month window; H3 is not yet testable and is deferred to the prospective version of the protocol. The prospective execution against the next MobilityData snapshot will be reported as a follow-up note in the same research programme.

This audit concerns the open specification itself, not the operators publishing under it. A2, A3, A6 and A7 detect patterns the GBFS specification *permits* (a constant placeholder, a conditionally-averaged virtual-station fleet, a zero-dock station, an empty capacity field). What our framework flags is that those permitted patterns are mutually incompatible with the arithmetic a downstream research consumer would naturally apply to the field. The technical lesson is about the standard, not about the publishers; criticism of the operators we name is out of scope for this paper. The standards-side recommendation in Section 5 is therefore phrased as a request for specification revision, not for operator sanction.

The audit is static: the `station_information` feed describes the deployment, not its dynamic behaviour. Real-time availability, rebalancing patterns and per-vehicle relocation live in the `station_status` feed, which carries its own quality issues and would require an analogous but disjoint audit protocol. The dynamic audit is the subject of the L2 limitation and the E4 future-work item.

5.3 When a clean verdict hides a missing column

The Italian *Dott* case documented in Section 3.4 illustrates what is at stake: a country that passes A1–A6 cleanly is not necessarily clean: it may be silent on the only field that matters for infrastructure quantification. An audit framework that conflates *no problem detected* with *no data published* mis-reads the entire national corpus. A7's empirical role is precisely to keep those two cases distinguishable.

5.4 GBFS v3 and the durability of the v2 legacy

GBFS v3 (released 2023) introduces `is_virtual_station`, a boolean that in principle disambiguates free-floating anchors from physical docks and would mechanically suppress A3 detection. The seven-class taxonomy nevertheless remains relevant under v3, for three reasons. Adoption is slow: of the 1,509 audited systems, only 541 (35.9 %) declare GBFS v3 in their auto-discovery feed at the audit date, against 772 (51.2 %) on v2.x; 50 remain on v1 and 146 expose no parseable version. The v2 legacy will persist: regulatory feeds on *transport.data.gouv.fr* and the MobilityData canonical catalogue archive snapshots that any longitudinal study from 2017 onward will need to ingest, and a v3 feed published today does not retro-fix the v2 historical record. And, most importantly, adoption is not compliance: the German cargo-bicycle mis-detection we report (Section 3.4) and the Italian Dott NaN-capacity propagation both occur on GBFS v3 feeds, because v3 makes the `is_virtual_station` flag *available* but does not make it *mandatory* or *verified*. The seven-class taxonomy A1–A7 therefore remains operationally relevant under full v3 adoption, and the protocol is straightforwardly extensible to the additional v3 attributes by adding rule-level checks at ingestion time.

5.5 Limitations

The protocol has five acknowledged limitations.

L1 – National coverage. The corpus is metropolitan France; extension to overseas territories and other jurisdictions requires revisiting the macro-regional surface threshold and the geofilter bounds.

L2 – Static versus dynamic. The audit is scoped to the static `station_information` document; auditing `station_status` (real-time availability) is a distinct problem. The salient risk is the *zombie station*: a physically dismantled station whose `station_information` entry has not been pruned leaves a ghost record that the A1–A7 detectors, operating on the published schema, certify as compliant. Consumers depending on station-presence should cross-check a recent `station_status` snapshot. A 2-day pilot (E4) already flagged candidate dynamic anomalies on 12.9 % of monitored stations; the collection pipeline ships with the repository and the formal dynamic extension is a planned follow-up.

L3 – Free-floating scope. Free-floating entries are audited as much as docked ones (A3 and A7 carry their capacity-publication anomalies), and the parquet exposes both `capacity_raw` and an audit-aware `capacity_audited` (null for free-floating). What v1.0 does *not* cover is their *dynamic* behaviour (rebalancing, real-time density), which belongs to the `station_status` audit of L2.

L4 – Operator coverage. Some operators publish partial feeds or are unavailable during the audit window, so inventory completeness is not guaranteed and must be re-evaluated periodically (experiment E3).

L5 – Scope and independence of the human check. The construct-validity check (Section 4.5) covers the rules verifiable from spatial imagery (A1, A3, A4; the v2 campaign shows that A5's declared-perimeter construct is only partially imagery-verifiable). A2 and A7 are structural data-property rules confirmed at the system level only (each is represented by a single operator in this catalogue snapshot), and A6 and the A3-boundary stratum had no eligible stations, so their per-station recall is not estimated. Most importantly, the check is *not independent of the authors*: the two annotators are this paper's authors (domain experts from the same institution), so it establishes that the detectors track physical reality, not that they were validated by a disinterested third party. The blinding (to the pipeline and to each other) limits anchoring but cannot remove this conflict; an external, multi-institution annotation is the necessary next step and is left to future work. Four conventions, compatible with the current GBFS specification, would close most of the anomalies documented here: populate `vehicle_type` exhaustively (A1); leave `capacity` as `null` on virtual stations rather than using placeholders (A2); adopt the GBFS v3 `vehicle_status` extension for free-floating fleets and stop publishing per-vehicle anchors as `station_information` (A3); publish geographical perimeters as polygons in the `geofencing_zones` block (A4, A5); and avoid sentinel coordinates $(0, 0)$ on stations whose geolocation is pending, instead omitting them from `station_information` until populated; a single such sentinel in Sevici Seville inflated its bounding-box area by four orders of magnitude in our E5 panel.

Methodological recommendation on A4 / A5 coupling. The Sevici finding (Table 11) suggests that the A5 bounding-box criterion be evaluated on the A4-filtered set of stations, not on the raw set. We retain the as-shipped definition in this paper to make the dependency explicit, but downstream implementations should consider an A5 variant computed after the A4 outlier removal step.

5.6 Future directions

Each limitation is paired with a falsifiable experiment (Table S10); six of the nine already include a completed pilot. The open priorities are the full threshold sweep (E2), the twelve-month temporal stability (E3) and the free-floating-native protocol (E6), with all protocols and scripts released under `experiments/`.

5.7 Downstream use

The Audit Catalogue unlocks four families of downstream studies: supply-side composite-index work on cycling-environment quality at station and commune resolution; demand modelling with graph-based neural networks [49] whose false-station nodes from A3 are a known confound; spatial-equity analysis combining station access with INSEE Filosofi; and multimodal integration with GTFS, for which the hybridisation schema is compatible with the canonical GTFS validator [17].

Data and code availability

Lead contact. Author contact details are withheld for double-blind review.

Materials availability. This study did not generate new physical materials. The released artefact is the *GBFS France Audit Catalogue* dataset, a derivative of public open data.

Data and code. The certified *GBFS France Audit Catalogue* (46,307 stations, 46 typed columns with per-row anomaly flags and audited capacities) is deposited on Zenodo under ODbL v1.0 (DOI `[deposited on Zenodo; DOI withheld for review]`, release 1.0.0; mirrored on *Recherche Data Gouv* and as a Hugging Face dataset at `[anonymised dataset mirror]`), alongside the rejected-stations log and machine-readable descriptors (JSON Schema, DCAT-AP, Frictionless Data Package, Croissant). The audit pipeline and a Docker image for bit-exact reproduction are released under the MIT licence at `[anonymised repository; link provided on acceptance]`; the A1–A7 detectors it calls are the `gbfs-toolkit` library (`pip install gbfs-toolkit`; MIT; source at `[anonymised library repository; link provided on acceptance]`), independently archived on Zenodo under DOI `[reference-implementation library archived on Zenodo; DOI withheld for review]`. The global-sweep snapshot underlying Section 3.4 is frozen under `experiments/.../global_audit_snapshot/` with a `verify_snapshot.py` that re-derives and asserts every reported global count; the blinded annotation sample, the robustness-check scripts and the sensitivity notebooks ship as supplementary material.

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CRediT authorship contribution statement

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Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

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